

Measuring What We Never Asked

Grouping survey items into population inferences, with the sample design carried through

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This project started with a battery and no principled way to summarize it

We field 11 items to measure something no respondent answers directly, and how we combine them has always been settled by convention rather than by evidence.

THE TWO DEFAULTS

- ▶ **Average them into an index.** Readable, but it assumes every respondent differs from every other only in *how much*.
- ▶ **Report all 11 separately.** Faithful, but no policymaker product can carry 11 lines, so someone collapses them anyway, off the record.

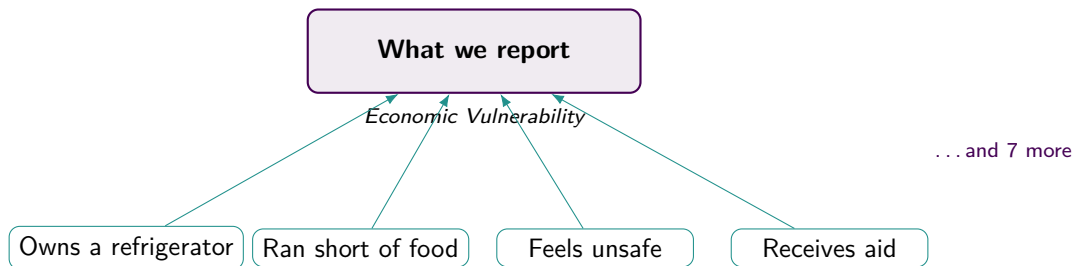
THE QUESTION NOBODY HAD TESTED

Is the index a summary of the battery, or an artifact of the decision to build one?

That is an empirical question. This project answers it, and carries the sample design through the answer.

Nobody in our data was ever asked the question we report on

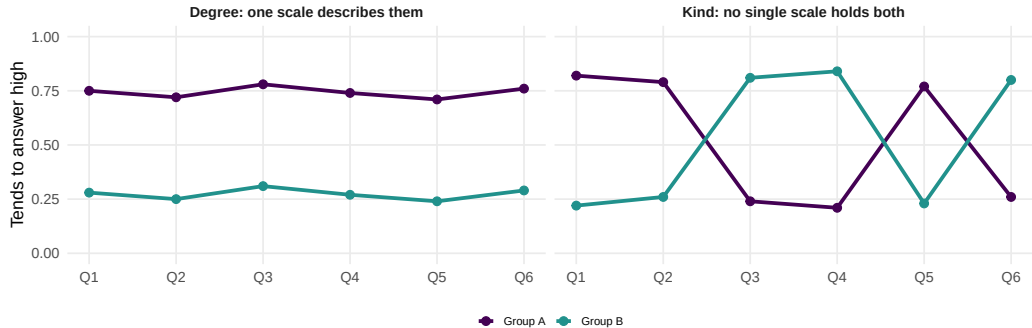
Every statement we publish about economic vulnerability is an inference from proxies, and its quality depends entirely on how those proxies are combined.



On what basis do we decide that these 11 answers add up to that one thing?

A battery can differ in degree or in kind, and only one summary is right

If people differ in how much, one scale describes them; if they differ in which answers they favor, that scale averages opposite people into a middle where nobody actually sits.



Which picture our battery matches is a question the data can answer. Until now we have answered it by assumption.

Roadmap

You have the question; here is the order in which the rest of this answers it.

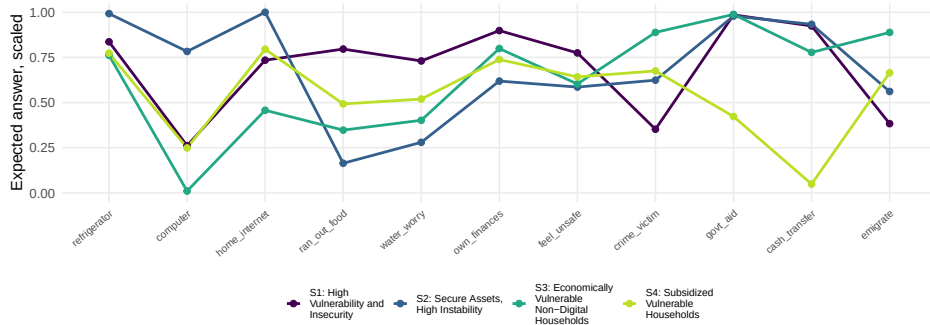
1. **What we found** on AmericasBarometer 2023, Ecuador
2. **How the method decides** degree versus kind
3. **Why the sample design** changes the answer
4. **Where AI fits**, and where it must not
5. **What I need from you**: how to capture this in production

The claim in one line

Grouping survey items into a population inference is a measurement decision and a design decision at once. Doing either alone gives a number we cannot defend.

On AmericasBarometer 2023, Ecuador, the answer is kind, not degree

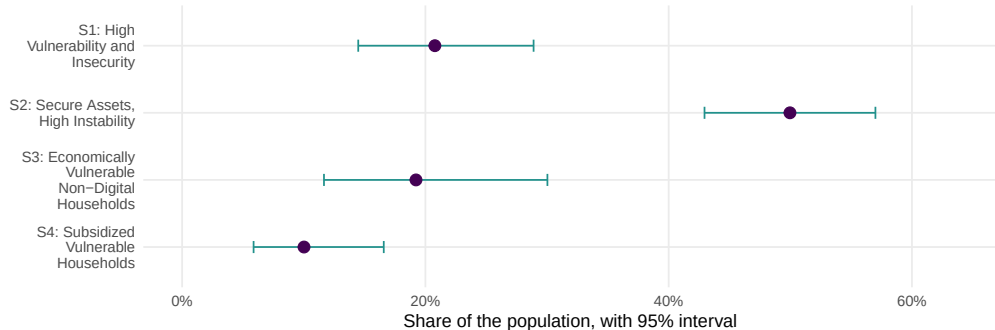
The battery does not line respondents up on one scale; it separates them into 4 recognizable groups, which means an index built from it has been averaging distinct publics together.



Lines that cross are the evidence: these groups reorder the items rather than sitting above and below each other.

These are population shares, not clusters that exist only in this file

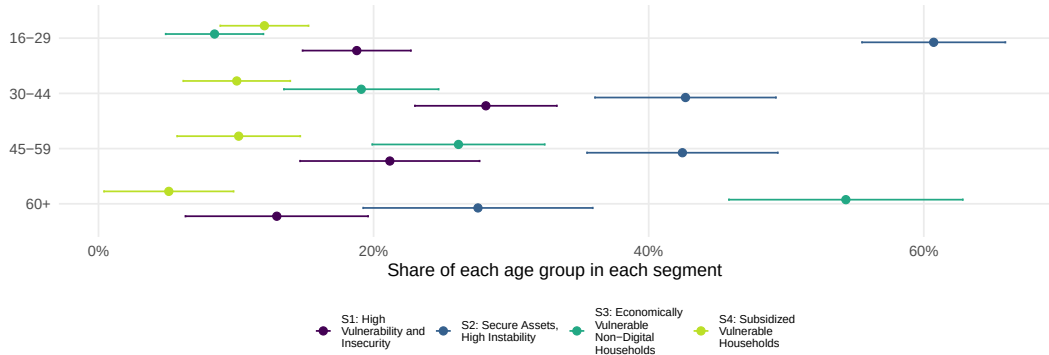
Each segment is estimated as a share of the adult population with its own margin of error, so these are numbers a product can carry.



The largest segment is 50% of adults (43 to 57%); the smallest is 10%. Intervals come from the sample design, not from a formula that assumes independent respondents.

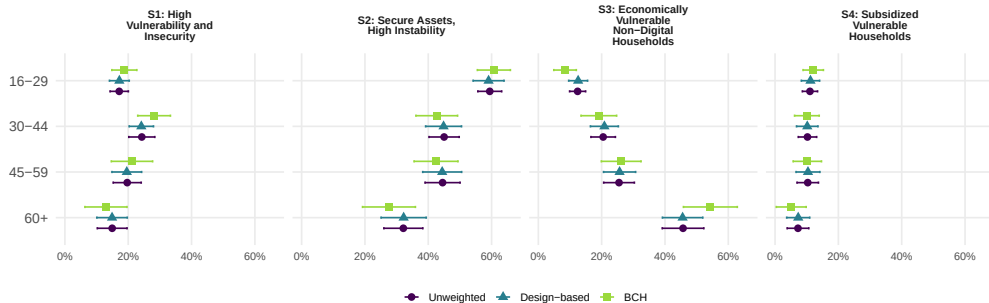
The segments cut across age group in ways the topline hide

Groups that look similar on a topline concentrate in different segments, which is why subgroup trends built on an index have looked flat.



What we would have reported, and what the data support

Reading across the three versions shows first what ignoring the sample design costs, then what forcing each respondent into one box costs, and the second gap is the larger one.



Unweighted is the plain cross-tab. **Design-based** adds the weights and the clustering. **Corrected** additionally undoes the shrinkage that assigning each respondent to one segment introduces.

The method groups respondents by pattern, not by score

A latent class model asks which answers travel together, so it can find people who are secure on some items and exposed on others, which a scale cannot represent.

WHAT THE MODEL ESTIMATES

- ▶ How many kinds of respondent there are
- ▶ For each kind, the probability of every answer to every item
- ▶ For each respondent, which kind they most resemble, and how sure we are

WHY IT SUITS US

Measurement error becomes a parameter we estimate rather than noise we hope cancels. Each item is treated as a fallible reading of something we cannot observe.

The number of groups is never chosen by the software. The analyst reads the evidence and records the decision.

k-means would have returned groups whether or not any existed

Clustering gives whatever number of groups we ask for, cannot use the survey weights, and reports no uncertainty, so it cannot tell us a segment is real or how large it is in the country.

	k-means	This method
How many groups	Whatever k you pass	Evidence plus a recorded analyst decision
Survey weights	Not used	Enter every estimate
Clustered sample	Ignored	Carried through every margin of error
Uncertainty	None reported	Interval on every quantity
Mixed item formats	Distance is arbitrary	Handled natively

Our samples are clustered and weighted, so default software understates uncertainty

Respondents arrive in neighborhood clusters of about 19 under a stratified design, and standard packages compute margins of error as though each had been drawn independently.

THIS DESIGN

1,604 respondents ■ 3 strata ■ 84 sampling points ■ 81 degrees of freedom

People in the same neighborhood answer alike, so 1,604 interviews carry less information than 1,604 independent ones. Every interval here is rebuilt 84 times, leaving out one sampling point each time.

ERRORS THIS ADDRESSES

- ▶ **Sampling error** estimated from the design we actually used
- ▶ **Measurement error** estimated as a parameter, not assumed away
- ▶ **Item nonresponse**: partial responders scored, not discarded
- ▶ **Classification error** introduced by the method, then corrected

Treating the measurement model as known understates our uncertainty

When we also rebuild the model itself inside every replicate, the margins of error roughly double, so intervals computed the standard way are optimistic and some differences do not survive.

Standard practice fits the measurement model once and treats it as fixed when estimating subgroup shares. Refitting it inside every replicate multiplies the margins of error by a median of **2.17** and by as much as **4.22**.

What this means for a product

Report the segment shares. Be slower to report that two subgroups differ, and check that the difference survives the wider interval before it reaches a briefing.

The language model names segments and never touches a number

An LLM drafts segment names and reads the subgroup tables under a fixed instrument, after every estimate is final, so a wrong name is a wording problem and never a statistical one.

WHAT IT SEES

- ▶ Response probabilities, one segment at a time
- ▶ The question wording
- ▶ Which subgroups the analysis resolved

WHAT IT NEVER SEES OR DOES

- ▶ No margins of error, no test statistics
- ▶ Never decides whether a difference is real
- ▶ Never combines probabilities
- ▶ Names freeze to a file the analyst edits

Names vary between runs, which is exactly why they are drafts and never enter a calculation.

The same pipeline runs on any country and any battery

Porting to a new account changes six things, all in one configuration file, and the method applies wherever respondents differ in which answers they favor rather than how much.

WHAT CHANGES IN A PORT

The file path, the item list, the nonresponse codes, the design variables, the demographic recodes, the number of groups.

Everything else, including every margin of error, is unchanged.

WHERE IT COULD GO NEXT

- ▶ A second country on the same battery
- ▶ A different topic domain
- ▶ The same country across rounds, to see whether a segment is growing

Which account wants to go first?

The details cannot go in a policymaker product, so where do they live?

A two-page product can carry a segment name and a share, but not the design, the corrections, or the caveats, and I want your read on how we capture that so it survives past any one analyst.

THREE OPTIONS

1. A standing **methods annex** per account that each product links to
2. A shared **archive**: configuration, diagnostics, and frozen segment names, citable per run
3. A **one-pager** that travels with each product: what the segments are, what the intervals mean, what the method cannot support

WHAT I NEED ANSWERED

- ▶ What must be preserved for someone to reproduce a published number in two years?
- ▶ Who reviews a segment name before it reaches a briefing?
- ▶ What is the minimum caveat that must appear in the product itself?

Backup

How the correction works

Assigning each respondent to their most likely segment misclassifies some of them, which pulls subgroup differences toward each other; the correction estimates that misclassification and undoes it.

The model tells us, for every respondent, the probability of belonging to each segment. Those probabilities imply a table: of everyone truly in segment A, what share gets assigned to A, to B, and so on. Inverting that table reweights the cross-tab so the differences come back to their unattenuated size. The reweighting can push a cell slightly outside 0 to 100 percent, which is a property of the correction rather than an error.

How many segments, and how we decided

K	Log-likelihood	Parameters	BIC	Entropy
2	-9889.0	29	19989.1	0.676
3	-9774.8	44	19869.7	0.669
4	-9698.7	59	19826.7	0.710
5	-9656.0	74	19850.5	0.678
6	-9627.0	89	19901.6	0.655
7	-9602.3	104	19961.4	0.691
8	-9579.4	119	20024.5	0.717
9	-9560.1	134	20095.1	0.768
10	-9541.0	149	20166.2	0.767

Lower BIC is better fit for the parameters spent; entropy near 1 means respondents are cleanly separated. Neither decides alone, and the software never picks. The analyst reads these with the profiles and records the choice.

Which items are doing the work

Item	Separation
cash_transfer	0.467
computer	0.388
ran_out_food	0.340
govt_aid	0.284
home_internet	0.281
crime_victim	0.277
emigrate	0.270
own_finances	0.266

Separation is the average probability gap between two segments on that item, from 0 to 1. An item near 0 is carrying nothing and is a candidate for removal.

Validated against an independent implementation

At equal weights our estimator and a published package must agree, and the pipeline checks this on every run rather than asserting it.

Largest gap in segment sizes: 0.00000. Largest gap in response probabilities: 0.00000. The run halts if these exceed a tolerance.